BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 320 · Final mixed-paper readiness

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Difference between MAR and MDR?

Q2. How to encode a negative integer in two's complement?

Q3. What does $O(n^2)$ describe?

Q4. Full adder A=1 B=1 carry-in=1. State sum and carry.

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I Do · Read new knowledge

Key vocabulary

readiness — Ability to apply without prompts.

synoptic — Knowledge across multiple topics.

method evidence — Reasoning supporting an answer.

New learning

Identify — Retrieve knowledge across both components

Model — Show working in numeric and algorithm tasks

Apply — Check terminology and boundary cases

Evaluate — Prioritise a final independent improvement target

Worked example

Normalise positive $0.0101_2 \times 2^3$.

1. Explain the method — Shift mantissa one position left and reduce exponent by one.

We Do · Practise with guidance

Search 9 in [1,3,5,7,9] using binary search.

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mid2→5, low3; mid3→7, low4; mid4→9 found.
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Explain the method

Each midpoint comparison reduces the interval; return index4.

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Q1 [8]: In a simplified sequential fetch, PC=12 and memory address 12 stores instruction ADD 7. State MAR, MDR, CIR and next PC (assume no branch or interrupt). [4] Convert −5 to 8-bit two's complement. [4]

Question 2

Q2 [8]: Trace binary search for 11 in [1,3,5,7,9,11,13]. [4] Explain base case and recursion progress. [4]

Question 3

Q3 [9]: Evaluate cloud storage versus USB-only storage for an inclusive school revision service; weigh access, reliability, security. [9]

Exit Ticket · Answer before checking

Publishing every lesson guarantees each student is ready for assessment.

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Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. MAR holds address; MDR transports data/instruction contents.

Q2. Invert fixed-width positive representation then add one.

Q3. Quadratic asymptotic growth.

Q4. 1+1+1=3=11_2; sum1 carry1.

You Do pages · Independent questions

Q1. + PC value 12 copied into MAR=12 [1]; MDR receives ADD 7 from memory location 12 [1]; CIR receives ADD 7 from MDR [1]; PC advances to 13 [1]; +5=00000101 [1], invert11111010 [1], add1 11111011 [2].

Q2. mid3→7 [1], low4 [1], mid5→11 [1], return5 [1]; base case stops [2], input progresses [1] avoiding infinite stack growth [1].

Q3. Indicative content: availability across devices, offline access, connectivity, inclusive accessibility, confidentiality, school account permissions, backup and recovery, and cost. Practice marking (authored guidance, not an OCR mark scheme): 7–9 = accurate, sustained evaluation of both options, applied to this scenario with a supported judgement; 4–6 = relevant comparison with some development and a partly justified decision; 1–3 = basic statements, limited context or unsupported decision; 0 = no creditworthy response. Use best-fit and credit equivalent valid arguments.
Model evaluation: Cloud storage gives pupils access to revision materials across school and home devices and helps teachers distribute updates without copying files manually. However, some pupils have unreliable internet or limited access to suitable devices, and any outage can interrupt revision. USB-only storage works without a connection but is easy to lose, can be outdated and may create a safeguarding or data-security risk if sensitive content is included. For an inclusive revision service I would use an access-controlled cloud platform as the main source, while providing approved offline downloads or school-based access where needed. Accessibility settings, restricted permissions, tested backups and a procedure for updating offline copies should be part of the design. The best decision depends on actual pupil connectivity and devices.

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Exit Ticket · Model answer

False. Independent performance and personalised revision evidence matter.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: